

QF632 Financial Data Science — Group Project

Milestone 2 · Preprocessing & Exploratory Data Analysis

Group 4 · 6 June 2026

1. Dataset & Pipeline Overview

We test whether the pricing linkage between European (TTF), U.S. (Henry Hub) and Asian (JKM) natural-gas benchmarks — normally disciplined by LNG arbitrage (Law of One Price) — becomes regime-dependent under geopolitical supply stress. This milestone documents the data engineering behind that test: a reproducible ingestion layer, a point-in-time (PIT) alignment that prevents look-ahead bias, and the missing-data and scaling decisions that make the mosaic model-ready.

All series were re-acquired from public sources over 2019-01-02 → 2026-06-05 and reconciled row-for-row against the Milestone 1 proof of ingestion (Appendix A). The aligned master spans 1,870 gas trading days.

Source family	Series (M1 columns)	Freq.	Access	Coverage
Market (Yahoo / yfinance)	HH, TTF, JKM, Brent, EUR/USD	daily	keyless	2019–2026 · exact
Macro (FRED)	US INDPRO; Euro IP	monthly	keyless	US→2026; euro→2024*
US storage (EIA)	Lower-48 working gas	weekly	keyless	2019–2026
EU storage (GIE AGSI+)	DE % full; TWh	daily	x-key	2019–2026 · exact
EU LNG (GIE ALSI)	ES send-out; inventory	daily	x-key	2019–2026 · exact
Weather (Open-Meteo ERA5)	HDD AMS / CHI / BEI	daily	keyless	2019–2026
Risk (Caldara–Iacoviello)	GPR / GPRT / GPRA / GPRC_RUS	monthly	keyless	2019–2026 · exact
Risk (policyuncertainty)	EU EPU	monthly	keyless	2019–2026

**Euro IP (OECD MEI series) was discontinued in 2024; it is retained but reverts to NaN thereafter (staleness rule, §2). Russian→EU pipeline flow (ENTSOG) is deferred to Milestone 3 (point-key resolution pending) — it is highly collinear with the war-regime indicator and the EU-storage drawdown, so M2 does not depend on it.*

2. Time-Series Structural Alignment

Master calendar. The decision grid is the union of gas trading days. Every observation carries two dates: event_date (the period it describes) and available_at (when it was first publishable in reality). All joins use pandas.merge_asof(direction='backward') keyed on available_at — never on event_date — so a value enters the grid only once a market participant could actually have known it.

Publication lags. Each series carries a conservative (upper-bound) release rule and a frequency-based staleness tolerance, beyond which a discontinued series reverts to NaN rather than being carried forward indefinitely:

Series	Freq.	available_at rule	Stale tol.
market	daily (trading)	T+0 exchange settle (known at close of day)	7 d
us_storage	weekly	EIA weekly report: week-ending Fri released following Thu ($\approx T+6$)	14 d
de_storage	daily (gas-day)	GIE AGSI+ gas-day published next morning (T+1)	10 d
es_lng	daily (gas-day)	GIE ALSI gas-day published next morning (T+1)	10 d
hdd	daily	Open-Meteo ERA5 reanalysis lag $\approx T+5$	14 d
us_ip	monthly	FRED INDPRO released mid next month (conservative month-end +20d)	45 d
euro_ip	monthly	Eurostat euro-area IP released $\approx$ month-end +45d	45 d
gpr	monthly	Caldara-Iacoviello GPR updated $\approx$ 10th of next month	45 d
epu	monthly	EPU Europe published early next month ($\approx$ month-end +7d)	45 d

Leakage proof. A point-in-time snapshot on 2022-02-25 (first trading day after the Russian invasion) makes the lag structure explicit — on that day the model knew only January macro/risk values, not February's:

Series	Freq.	event_date	available_at	age (d)
market	daily (trading)	2022-02-25	2022-02-25	0
us_storage	weekly	2022-02-18	2022-02-24	7
de_storage	daily (gas-day)	2022-02-24	2022-02-25	1
es_lng	daily (gas-day)	2022-02-24	2022-02-25	1
hdd	daily	2022-02-20	2022-02-25	5
us_ip	monthly	2022-01-01	2022-02-20	55
euro_ip	monthly	2021-12-01	2022-02-14	86
gpr	monthly	2022-01-01	2022-02-10	55
epu	monthly	2022-01-01	2022-02-07	55

A unit-test suite (tests/test_alignment.py) asserts `available_at` $\leq$ cutoff and `event_date` $\leq$ cutoff for all nine series across all 1,870 days, that T+1 fundamentals are never used same-day, and that no series is stale-forwarded beyond tolerance — all five tests pass.

3. Scale Disparities & Missing Data

Missing data. Values are handled by mechanism, with provenance flags — never silently imputed:

Situation	Treatment	Rationale
Cross-market holidays (price/FX)	forward-fill ≤ 2 trading days, flag stale	short-horizon mean-reverting; 2–3 cells/series
Monthly/weekly before first release	leave NaN (no backfill)	back-filling would fabricate early signal
Discontinued series (euro IP)	NaN past tolerance window	avoid propagating a years-old value
ENTSOG flow = 0 (when added)	keep 0, flag zero	0 is a real pipeline shut-in, not missing

Outliers. Extremes are flagged, never winsorized — the 2020/2022/2026 regime extremes ARE the object of study. A rolling 30-day robust-MAD screen ($k=6$) writes 161 flags to outliers.csv; as a sanity check the invasion window (2022-02-24...03-15) surfaces TTF, JKM and both spreads. The screen also exposes JKM's low-liquidity quoting (flat runs punctuated by jumps), an operational caveat carried into the pitch.

Scaling. Dual-track and strictly causal. Levels (with spreads) live in master_clean.parquet; the modelling matrix master_scaled.parquet uses log-returns and a rolling 252-day z-score (min 126) — never full-sample statistics, which would leak the future. Bounded storage % uses a logit; LNG send-out uses log1p; index levels use YoY. TTF is converted to USD/MMBtu ($\times \text{EUR/USD} \div 3.41214$) so the spreads are unit-consistent.

4. Exploratory Findings

The cross-basin spreads collapse to near-zero in calm periods (Law of One Price) and blow out under supply stress: TTF–HH peaks near 90 USD/MMBtu during the 2022 energy crisis and re-widens during the 2026 Hormuz closure, while U.S. Henry Hub barely moves — it is insulated from the chokepoint.

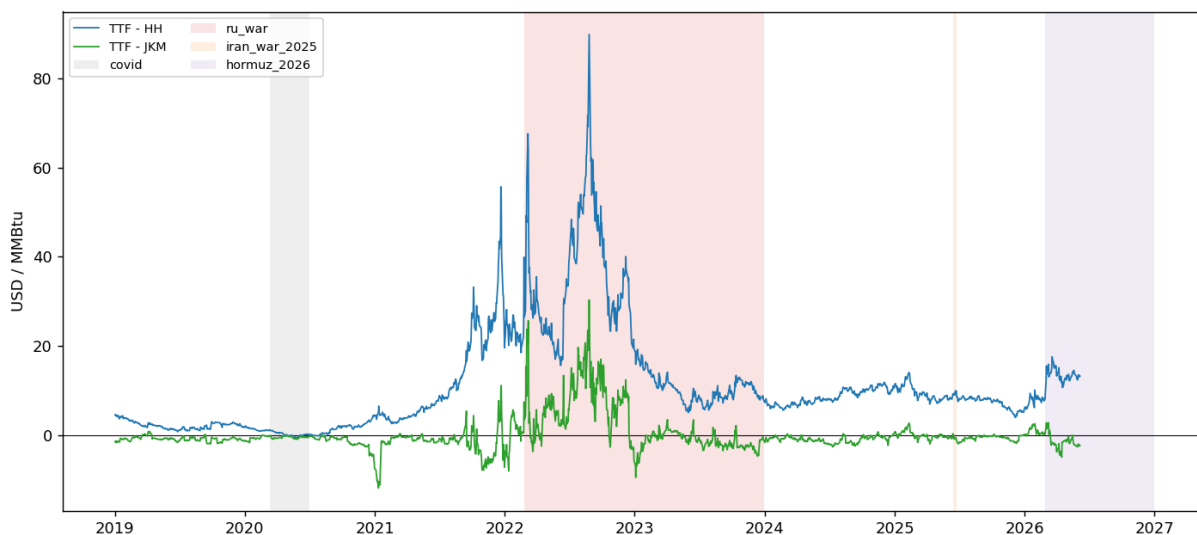


Figure 1. Cross-basin spreads (USD/MMBtu) with regime shading. Near-zero under Law-of-One-Price; TTF–HH blows out to ~90 in the 2022 war and re-widens in the 2026 Hormuz closure.

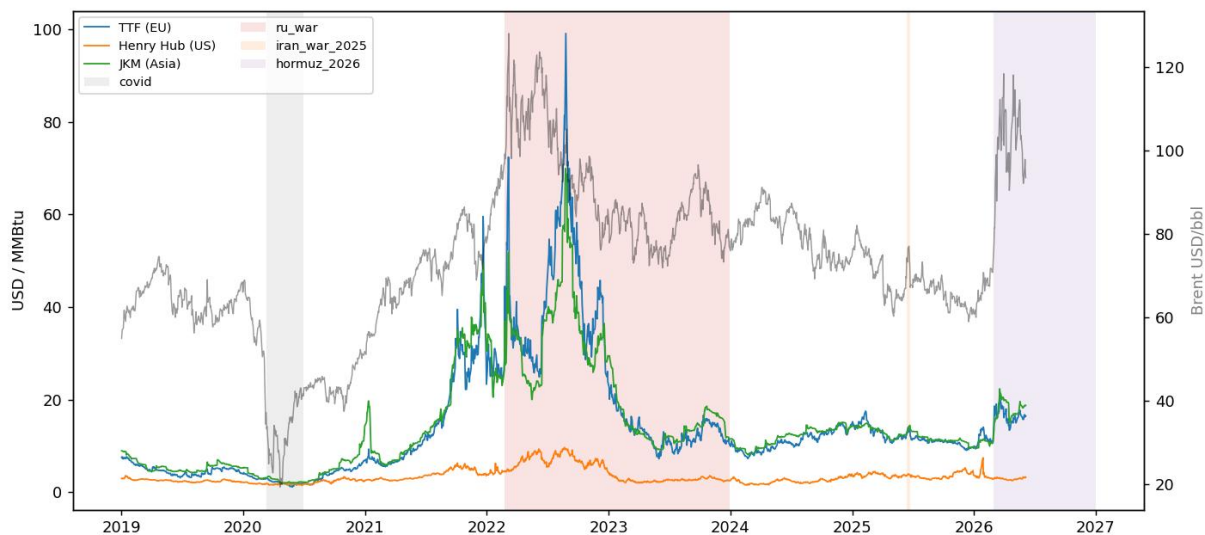


Figure 2. Benchmarks in common units, Brent on the right axis. Henry Hub stays low and stable while TTF and JKM spike together under stress.

Regime means confirm the structure: the European supply premium and Russia-specific geopolitical risk peak in the 2022 war, while the 2026 Hormuz closure lifts TTF, JKM and Brent together as Henry Hub stays flat.

regime	TTF-HH	TTF-JKM	TTF	HH	GPR	GPRC_RUS
normal	7.44	-0.88	10.46	3.02	109.45	1.17
covid	0.23	-0.43	1.98	1.74	74.17	0.45
ru_war	22.22	2.80	26.85	4.62	139.07	2.88
iran_war_2025	9.34	-0.65	13.09	3.75	166.05	1.49
hormuz_2026	13.63	-1.47	16.54	2.91	225.83	1.84

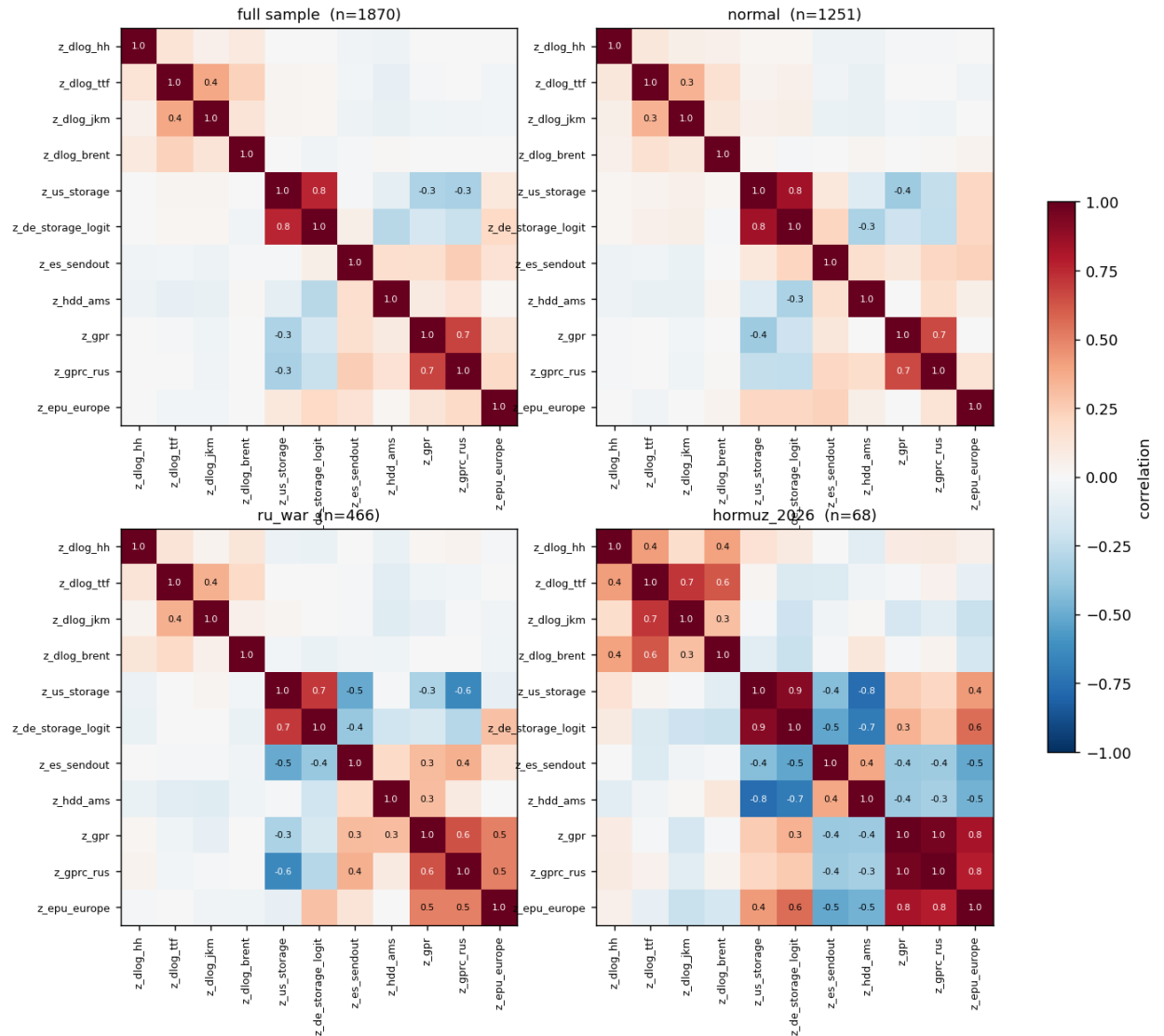


Figure 3. Correlation structure by regime ($|r| \geq 0.3$ annotated). Cross-basin co-movement reorganises as supply stress sets in.

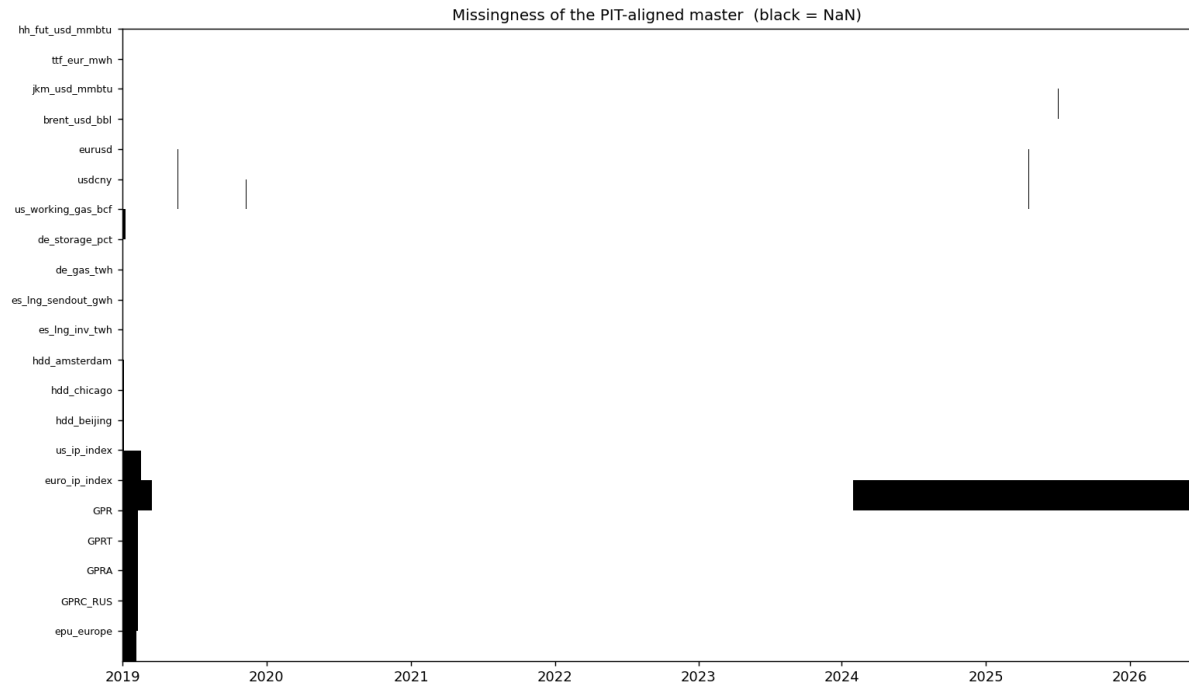


Figure 4. Missingness of the PIT-aligned master (black = NaN): pre-first-release gaps and the euro-IP discontinuation — no fabricated fills.

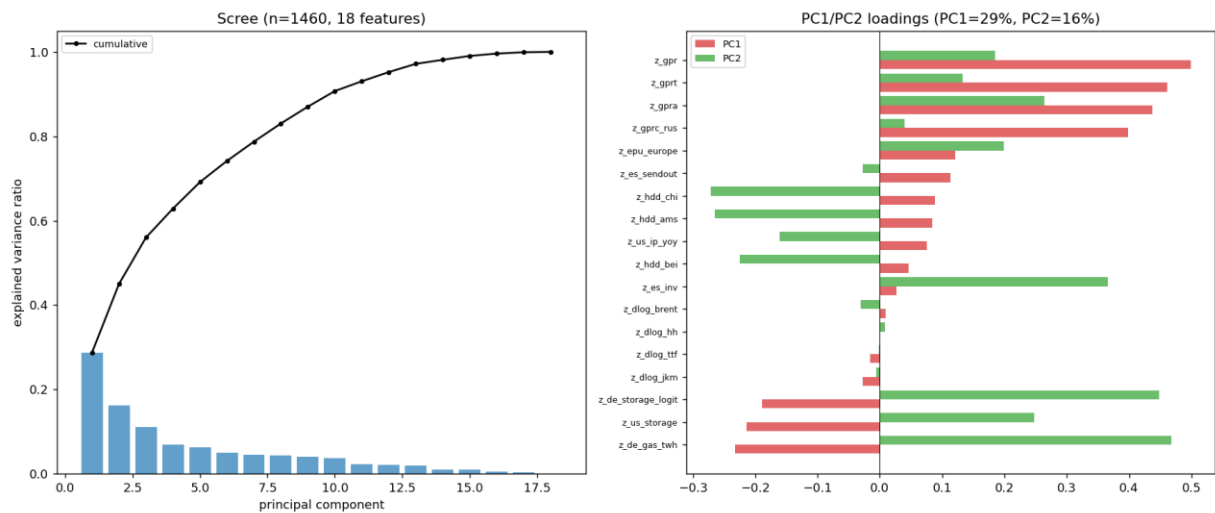


Figure 5. PCA preview (M3). PC1 (~29%) contrasts geopolitical risk with storage; the first three components explain ~56% of variance.

Stationarity (ADF + KPSS, reports/stationarity.csv): spread_ttf_hh is non-stationary in levels (persistent regime level-shifts), so Milestone 3 will difference it or condition on regime; spread_ttf_jkm and all z-scored daily features are stationary. Regime validation (validate_regimes.py): the structural break lands exactly on 2026-03-02 (TTF +39% in one day, Henry Hub -9%), confirming the Hormuz boundary data-drivenly.

5. Operational Notes (pitch readiness)

- JKM is thinly traded — stale quotes punctuated by jumps; any cross-basin strategy faces real execution friction and bid-ask cost on the Asian leg.
- Regimes are calibrated to verified event dates: COVID (control), 2022 Russia (positive control), the Jun-2025 Twelve-Day War (precursor — strait stayed open), and the Feb–Mar 2026 Hormuz closure (candidate chokepoint regime).
- The pipeline is reproducible end-to-end (ingest → align → clean → eda → report); API keys live in .env and are never committed.

Appendix A — Proof of Ingestion (first 5 rows)

Re-ingested from public sources; values match the Milestone 1 submission exactly.

Market — prices & FX (Yahoo)

date	hh_fut_usd_mmbtu	ttf_eur_mwh	jkm_usd_mmbtu	brent_usd_bbl	eurusd
2019-01-02	2.958	22.475	8.970	54.91	1.1462
2019-01-03	2.945	22.255	8.945	55.95	1.1318
2019-01-04	3.044	22.930	8.885	57.06	1.1391
2019-01-07	2.944	21.530	8.830	57.33	1.1410
2019-01-08	2.967	22.090	8.820	58.72	1.1480

Physical — EU gas storage, Germany (GIE AGSI+)

date	de_storage_pct	de_gas_twh
2019-01-01	75.23	189.1272
2019-01-02	75.01	188.5782
2019-01-03	74.47	187.2198
2019-01-04	74.10	186.2821
2019-01-05	73.96	185.9276

Physical — EU LNG send-out & inventory, Spain (ALSI)

date	es_lng_sendout_gwh	es_lng_inv_twh
2019-01-01	338.7	2259.35
2019-01-02	461.2	2297.99
2019-01-03	538.6	2229.83
2019-01-04	547.1	2143.05
2019-01-05	407.0	2359.09

Physical — US Lower-48 weekly storage (EIA)

date	us_working_gas_bcf
2019-01-04	2614
2019-01-11	2533
2019-01-18	2370
2019-01-25	2197
2019-02-01	1960

Physical — Weather HDD (Open-Meteo)

date	hdd_amsterdam	hdd_chicago	hdd_beijing
2019-01-01	10.1	18.0	23.0
2019-01-02	12.1	19.8	23.3
2019-01-03	13.1	19.5	21.5
2019-01-04	12.3	16.1	21.9
2019-01-05	10.5	15.1	22.2

Risk — Geopolitical Risk incl. Russia (Caldara-Iacoviello)

date	GPR	GPRT	GPRA	GPCR_RUS
2019-01-01	87.4241	112.9805	55.7335	0.9987
2019-02-01	96.8002	127.8574	57.5117	1.0575
2019-03-01	82.3283	90.7896	66.6412	0.5631
2019-04-01	79.2534	95.4554	57.5155	0.8058
2019-05-01	106.4719	134.9725	64.6634	0.9714

Risk — European EPU (policyuncertainty)

date	epu_europe
2019-01-01	247.6771
2019-02-01	230.1516
2019-03-01	229.3842
2019-04-01	214.2044
2019-05-01	196.3067

Macro — industrial production (FRED)

date	us_ip_index	euro_ip_index
2019-01-01	103.4021	106.4
2019-02-01	102.8371	106.4
2019-03-01	102.8757	106.6
2019-04-01	102.2740	106.1
2019-05-01	102.3920	106.4